

James R. Bounds

Curriculum Vitae

College Station, Texas · jamesbounds@tamu.edu

EDUCATION & EMPLOYMENT HISTORY

- 2015** **B.S. Physics**
Texas A&M University
Department of Physics and Astronomy
- 2012–2015** **Undergraduate Research Assistant**
Texas A&M University
Advisor: Prof. Hans Schuessler
- 2016** **IT Technician**
Systek Computing
College Station, Texas
- 2016–2025** **Graduate Research Assistant**
Texas A&M University
Advisor: Prof. Hans Schuessler
- 2025–2026** **Graduate Teaching Assistant**
Texas A&M University
Advisor: Prof. Hans Schuessler, Prof. Alexei Sokolov

RESEARCH EXPERIENCE

- Design and construction of high precision scientific instruments
- Precision laser spectroscopy and cavity-enhanced absorption techniques
- Characterization of breath samples using laser spectroscopy
- High-resolution measurement of molecular absorption cross section
- Retrieval of hyperfine constants from doppler-free spectra
- Demonstration of novel wavelength and optical frequency calibration techniques
- Optical system design and alignment (CRDS, Vernier spectroscopy, interferometry)
- Scientific instrument design (high precision spectrometers and wavemeters, novel measurement platforms)
- Scientific software development (full-stack, numerical optimization, instrument automation, GUI design)
- Electronics and instrumentation (DAQ systems, photodetectors, control electronics)
- Experimental characterization, uncertainty analysis, and validation against molecular databases

PUBLICATIONS

Peer Reviewed

- 2025** **Acetone Absorption Cross-Section in the Near-Infrared of the Methyl Stretch Overtone and Application for Analysis of Human Breath.**
Bounds, J., E. Aluaee, A. Kolomenskii, and H. Schuessler.
Optics **6** (1)
, URL: <https://www.mdpi.com/2673-3269/6/1/9>.
- 2023** **Hyperfine structure and isotope shifts of xenon measured for near-infrared transitions with Doppler-free saturated absorption spectroscopy.**
Bounds, J., A. Kolomenskii, R. Trainham, M. Manard, et al.
Spectrochimica Acta Part B: Atomic Spectroscopy **202**, p. 106635
URL: <https://www.sciencedirect.com/science/article/pii/S0584854723000228>.

- 2022 Spectroscopic trace gas detection in air-based gas mixtures: Some methods and applications for breath analysis and environmental monitoring.**
Xia, J., F. Zhu, J. Bounds, E. Aluauee, et al.
Journal of Applied Physics **131** (22), p. 220901
URL: <https://doi.org/10.1063/5.0091263>.
- 2019 Sensitive acetone detection with a mid-IR interband cascade laser and wavelength modulation spectroscopy.**
Xia, J., F. Zhu, A. A. Kolomenskii, J. Bounds, et al.
OSA Continuum **2** (3), pp. 640–654
URL: <https://opg.optica.org/osac/abstract.cfm?URI=osac-2-3-640>.
- 2018 Sensitive Spectroscopic Analysis of Biomarkers in Exhaled Breath.**
Bicer, A., J. Bounds, F. Zhu, A. A. Kolomenskii, et al.
International Journal of Thermophysics **39** (6), p. 69
URL: <https://doi.org/10.1007/s10765-018-2389-9>.
- 2017 Single ion fluorescence excited with a single mode of an UV frequency comb.**
Ozawa, A., J. Davila-Rodriguez, J. R. Bounds, H. A. Schuessler, et al.
Nature Communications **8** (1), p. 44
URL: <https://doi.org/10.1038/s41467-017-00067-9>.
- 2017 Probing methane in air with a midinfrared frequency comb source.**
Zhu, F., J. Xia, A. Bicer, J. Bounds, et al.
Appl. Opt. **56** (22), pp. 6311–6316
URL: <https://opg.optica.org/ao/abstract.cfm?URI=ao-56-22-6311>.
- 2015 Mid-infrared dual frequency comb spectroscopy based on fiber lasers for the detection of methane in ambient air.**
Zhu, F., A. Bicer, R. Askar, J. Bounds, et al.
Laser Physics Letters **12** (9), p. 095701
URL: <https://doi.org/10.1088/1612-2011/12/9/095701>.
- 2014 Near infrared frequency comb vernier spectrometer for broadband trace gas detection.**
Zhu, F., J. Bounds, A. Bicer, J. Strohaber, et al.
Opt. Express **22** (19), pp. 23026–23033
URL: <https://opg.optica.org/oe/abstract.cfm?URI=oe-22-19-23026>.

Preprint

- 2026 *Carrier Revival in Long Trapped-Ion Chains.***
Egli, F., C. Shanks, J. Bounds, J. Moreno, et al.
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CONFERENCE PROCEEDINGS

- 2023 Hyperfine Structure of Xe by Frequency Comb Calibrated Saturation and Collinear Fast Beam Spectroscopies**
31st Annual International workshop, Laser Physics 2023
H.Schuessler, J. Bounds, N. Schnoor, A. Kolomenskii, R. Trainham, M. Manard.
- 2023 Local and Remote Measurements of Atmospheric Trace Gases**
31st Annual International workshop, Laser Physics 2023
A. Kolomenskii, C. Rodriguez, J. Bounds, E. Aluauee, H. Schuessler.
- 2021 Broadband Frequency Comb Vernier Spectroscopy In The Mid-IR With Video-recording Retrieval Of Absorption Spectrum**
International Symposium on Molecular Spectroscopy
H. Schuessler, J. Bounds, A. Kolomenskii
DOI: [10.15278/isms.2021.FF12](https://doi.org/10.15278/isms.2021.FF12)

2021 Spectroscopic Detection Of Methane at a ppt Sensitivity Level in Mid-IR with a Long-path Multipass Cell
International Symposium on Molecular Spectroscopy
J. Xia, C. Rodriguez, J. Bounds, A. Kolomenskii, H. Schuessler
DOI: [10.15278/isms.2021.RF10](https://doi.org/10.15278/isms.2021.RF10)

2018 Frequency comb vernier spectroscopy of methane in the mid-IR with temporal retrieval of comb lines
International Symposium on Molecular Spectroscopy
James R Bounds, F. Zhu, A. Kolomenskii, H. Schuessler.
DOI: [10.15278/isms.2018.WA05](https://doi.org/10.15278/isms.2018.WA05)

2018 Acetone and methane detection with wavelength modulation spectroscopy in the near- and mid-IR
International Symposium on Molecular Spectroscopy
J. Xia, F. Zhu, J. Bounds, A. Kolomenskii, H. Schuessler.
DOI: [10.15278/isms.2018.RH03](https://doi.org/10.15278/isms.2018.RH03)

POSTER PRESENTATIONS

2025 Hyperfine structure and isotope shifts of xenon measured for near-infrared transitions with Doppler-free saturated absorption spectroscopy
SIBOR Workshop
Cook's Branch Conservancy, Montgomery, Texas

ORAL PRESENTATIONS

2024 Frequency Comb Calibration of the Hyperfine Structure of Metastable Xenon
SIBOR Workshop
Cook's Branch Conservancy, Montgomery, Texas

2023 Frequency Comb Calibration of the Hyperfine Structure of Metastable Xenon
SIBOR Workshop
Cook's Branch Conservancy, Montgomery, Texas

2022 Metastable Xenon Hyperfine Structure Measurement
SIBOR Workshop
Cook's Branch Conservancy, Montgomery, Texas

2019 Vernier Spectroscopy and Spectroscopic Detection of Biomarkers in Exhaled Breath
Workshop of Max Planck Institute of Quantum Optics Laser Spectroscopy Division
Schloss Ringberg, Kreuth, Germany

OPEN SOURCE PROJECTS

Github: <https://github.com/jaymz07>

- **Battery-Powered, Amplified Photodetector**
<https://github.com/jaymz07/Photodetector-Battery>
- **Quantum Tunneling Demo Written in Java**
<https://github.com/jaymz07/quantum-tunneling>
- **Gaussian Beam Simulator GUI**
<https://github.com/jaymz07/Gaussian-Beam-Sim-Qt>
- **Grotrian Diagram Generator**
<https://github.com/jaymz07/Grotrian>

TECHNICAL SKILLS

Programming

Python, Java, C, Cython, Verilog, C++, MATLAB, Octave, Mathematica, LabVIEW, Bash, PHP, git, network proficiency

Scientific Computing

NumPy, SciPy, PyTorch, Numba, openCV, Scikit-learn, Pandas, Slurm

Scientific Visualization

Matplotlib, Datashader, HoloViews, GNUPlot, Blender, PyVista, OpenGL, Origin, LaTeX

Instrumentation & Data Acquisition

Communication protocol proficiency (GPIB, RS232, USB, Ethernet, SPI, I²C), Instrument Automation (PyVISA, FPGA), GUI Design (QT, Tkinter, LabView)

Optics and Photonics

Laser alignment, optical fiber splicing, cavity-enhanced absorption spectroscopy, optical frequency combs, fiber optics, interferometry, precision wavelength metrology, optical frequency metrology, Fourier-transform spectroscopy, dual-comb spectroscopy, multi-pass cells

Electronics

Analog, digital, and mixed circuit design/simulation, PCB design, soldering, microcontrollers, FPGA, signal conditioning, power supply design, amplifier design, filter design, photodetector optimization, vector network analysis, PID controllers, ADC/DAC systems, embedded systems, single-board computers

Mechanical

CAD, precision machining (vertical mill and precision lathe), 3D printing, vacuum systems

Operating Systems

Linux (Debian, Ubuntu, MX Linux, openSUSE, Fedora, CentOS, Arch), Windows

Software Proficiency

Server configuration (openSSH, Apache, PHP, X11), 3D modeling (Autodesk Inventor, Blender, pyvista, OpenGL), Electronic circuit simulation (National Instruments Multisim, QUCS), PCB Design (KiCAD), Terminal Proficiency (Bash, zsh, vim, tmux), Image Editing (Adobe Photoshop, GIMP, Inkscape), Audio editing (Audacity, Ableton, FL Studio), presentation (LibreOffice Impress, Microsoft Powerpoint), spreadsheet (LibreOffice Calc, Microsoft Excel), L^AT_EX (this document), Web Page Design, git, Microsoft Word